

CTIM — Control Traceability Integrity Module

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Accountability Governance Architecture (AGA™)

Operational Layer: Control Governance Layer

Governed Space: Control Traceability Integrity

Category: Governance & Enforcement

Subcategory: Chain of Control Governance

Type: Chain of Control Standard Module

Parent Standard: Chain of Control Standard (CCS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 14 June 2026

Compatibility: OOF Methodology OS · Chain of Control Standard (CCS)

· Responsibility Governance Standard (RGS) ·

Evidence Accountability Standard (EAS) · Capability Readiness

Governance Standard (CRGS) · INTEGROS® —

Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Control Traceability Integrity Module (CTIM) defines the structural conditions under which operational control relationships, supervision activities, command decisions, control transfers, runtime control events, execution directives, and control outcomes remain traceable, reconstructable, reviewable, auditable, governable, and operationally valid throughout the control lifecycle.

CTIM governs control traceability.

The module establishes the integrity conditions required to determine who controlled operational reality, when control existed, how control changed, how control moved between actors, and whether the complete chain of control can be reconstructed after execution.

Module Operational Role

CTIM defines the control-traceability governance layer of CCS.

Its role is to preserve visibility and reconstructability of control relationships throughout operational reality.

Module Operational Space

CTIM governs:

- control traceability
- supervision traceability
- control reconstruction
- control auditability
- control-transfer traceability
- command-history traceability
- runtime-control traceability
- governance traceability

The module applies wherever governance requires reconstruction of control.

Module Function

The module applies wherever systems must preserve:

- traceable control relationships
- reconstructable supervision history
- auditable control decisions
- accountable control transfers
- governance-valid control records
- operational-control transparency

Its function is to ensure that governance can reconstruct who controlled operational reality throughout execution.

Minimum Implementation Framework

1. Define the Control Traceability Object

The organization must define which control environments require traceability governance.

This may include:

- management structures
- contractor environments
- subcontractor chains
- operational execution systems
- industrial operations
- compliance systems
- AI governance systems
- Human-AI operational systems

2. Define Control Traceability Conditions

The system must define the conditions under which control traceability remains valid.

This includes:

- traceability requirements
- reconstruction requirements
- evidence requirements
- auditability requirements
- accountability requirements
- governance-valid traceability conditions

3. Define Traceability Degradation Detection Logic

The system must define how control-traceability failures are identified.

This may include:

- missing supervision records
- undocumented control transfers
- incomplete command history
- evidence gaps
- governance-blind control activities
- unreconstructable control chains

4. Define Operational Response or Governance Logic

The system must define governance logic for control-traceability failures.

Governance response may include:

- control review
- audit procedures
- governance intervention
- control reconstruction
- corrective actions
- supervision verification
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- control assignments
- supervision activities
- command decisions
- control transfers
- governance reviews

- resulting control states

A control-governance environment must not remain traceability-valid if materially significant control activities cannot be reconstructed, reviewed, audited, validated, or governed.

Use Case 1 — Multi-Layer Contractor

Environment

Scenario

A project involves an investor, contractor, subcontractors, supervisors, and operational workers performing activities across multiple organizational layers.

Application

CTIM reconstructs who exercised control, when control existed, how supervision occurred, and how control moved throughout the operational chain.

Result

Governance gains visibility into the complete chain of control.

Use Case 2 — Multi-Agent AI Environment

Scenario

An orchestration platform coordinates multiple AI agents, delegates operational decisions, and dynamically transfers runtime control during execution.

Application

CTIM reconstructs control relationships, runtime control transfers, command decisions, and supervision history.

Result

Governance gains visibility into control pathways across intelligent operational systems.

Canonical Closing Statement

Control Traceability Integrity Module (CTIM) defines the structural conditions under which operational control relationships, supervision activities, command decisions, control transfers, runtime control events, execution directives, and control outcomes remain traceable, reconstructable, reviewable, auditable, governable, and operationally valid throughout the

control lifecycle.

Control that cannot be reconstructed cannot be governed. Control traceability integrity therefore becomes a foundational condition of trustworthy chain-of-control governance.

Related Documents

→ Chain of Control Standard - (CCS).

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Canonical Source

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